INTERNATIONAL MONETARY FUND

SENEGAL

December 18, 2020

SECOND REVIEW UNDER THE POLICY COORDINATION INSTRUMENT AND REQUEST FOR MODIFICATION OF QUANTITATIVE TARGETS—DEBT SUSTAINABILITY ANALYSIS UPDATE[1]

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Prepared by the staffs of the International Monetary Fund and the International Development Association

Joint Bank-Fund Debt Sustainability Analysis	
Risk of external debt distress:	*Moderate*[2]
Overall risk of debt distress	*Moderate*
Granularity in the risk rating	*Limited scope to absorb shocks*
Application of judgment	*No*

This Debt Sustainability Analysis (DSA) update indicates that Senegal's debt remains sustainable with a moderate risk of overall debt distress (external and public) under the current baseline, with limited scope to absorb shocks in the near term. This is in line with the previous DSA update of April 2020 that accompanied Senegal's RCF/RFI request. The moderate risk of debt distress is predicated on a gradual recovery from the pandemic-induced economic shock, continued adherence to a prudent fiscal path (in line with regional convergence criteria), and sound debt management. The outlook for Senegal remains highly uncertain and subject to significant downside risks, primarily a reflection of broader global risks related to the COVID-19 pandemic. A prudent borrowing strategy that prioritizes concessional external borrowing and domestic regional financing in line with programmed financing needs, combined with continued efforts to strengthen debt management and contain fiscal risks, will be the anchor for continued debt sustainability.

[1] This DSA was prepared jointly with the World Bank and in collaboration with the Senegalese authorities.

[2] Senegal's debt carrying capacity is classified as strong (with a CI score of 3.19 calculated based on the October 2020 WEO and the 2019 CPIA). The applicable thresholds to public and publicly guaranteed external debt are: 55 percent for the PV of debt-to-GDP ratio, 240 percent for the PV of debt-to-exports ratio, 21 percent for the debt service-to-exports ratio, and 23 percent for the debt service-to-revenue ratio. The applicable benchmark for the PV of total public debt for strong debt carrying capacity is 70 percent of GDP.

DEBT COVERAGE

1. This DSA uses a broad definition of public debt. As in the previous assessment, this DSA captures (i) the central government, (ii) para-public entities, and (iii) state-owned enterprises (SOEs), combining data provided by the authorities with information compiled by IMF technical assistance.[3]

Text Table 1. Senegal: Public Debt Coverage and Design of Contingent Liability Test

	Subsectors of the public sector	Sub-sectors covered
1	Central government	X
2	State and local government	X
3	Other elements in the general government	
4	o/w: Social security fund	
5	o/w: Extra budgetary funds (EBFs)	X
6	Guarantees (to other entities in the public and private sector, including to SOEs)	X
7	Central bank (borrowed on behalf of the government)	X
8	Non-guaranteed SOE debt	X

1	The country's coverage of public debt	The central, state, and local governments plus extra budgetary funds, central bank, government-guaranteed debt, non-guaranteed SOE debt		
		Default	Used for the analysis	Reasons for deviations from the default settings
2	Other elements of the general government not captured in 1.	0 percent of GDP	0.0	
3	SoE's debt (guaranteed and not guaranteed by the government) 1/	2 percent of GDP	0.0	
4	PPP	35 percent of PPP stock	2.1	PPP capital stock of 6 percetn of GDP is larger than 3 percent threshold
5	Financial market (the default value of 5 percent of GDP is the minimum value)	5 percent of GDP	5.0	
	Total (2+3+4+5) (in percent of GDP)		7.1	

1/ The default shock of 2% of GDP will be triggered for countries whose government-guaranteed debt is not fully captured under the country's public debt definition (1.). If it is already included in the government debt (1.) and risks associated with SoE's debt not guaranteed by the government is assessed to be negligible, a country team may reduce this to 0%.

BASELINE SCENARIO

2. This DSA updates the April 2020 assessment which had incorporated the initial impact of the COVID-19 pandemic into the baseline. The April DSA treated the pandemic as a one-off large shock leading to a significant decline in GDP and a widening of fiscal and current account deficits. This DSA uses the same debt stocks for 2018 and 2019 as the previous assessment but incorporates revised revenue and expenditure data on para-public entities and state-owned enterprises (SOEs) and updates the macroeconomic outlook using staff's latest baseline projections. Compared to the previous DSA, the economic downturn is more severe in 2020 and the recovery more gradual, reflecting a deeper and more long-lasting shock than initially anticipated.

3. The revised baseline macroeconomic assumptions are the following:

- **GDP growth** is projected to be -0.7 in 2020 (3 percent in the previous DSA). The economy is assumed to gradually recover starting in 2021 with a growth rate of 5.2 percent (5.5 percent in the previous DSA) and remaining high over the medium term. Long-run GDP growth projections are broadly unchanged.
- Staff estimates that the overall ***public sector deficit*** will be 7.4 percent of GDP in 2020, higher

[3] The list of SOEs covered by the DSA is provided in the Technical Memorandum of Understanding.

than the 6.8 percent of GDP anticipated at the time of the RCF/RFI. This revision reflects the larger central government fiscal deficit resulting from the crisis. This figure also reflects updated revenue and expenditure data for para-public entities and SOEs through 2018 obtained from recent IMF technical assistance missions and revisions to PETROSEN's oil and gas investment schedule to reflect a slower investment schedule due to COVID-19. The latter two revisions were net positive for the 2020 public sector deficit and somewhat offset the decline in the central government balance.

- The ***current account deficit*** is projected to be 10.9 percent of GDP in 2020 and 10.7 percent in 2021. Although larger than projected pre-crisis, this is a smaller deficit than anticipated at the time of the April DSA (11.3 percent and 11.4 percent, respectively). The difference primarily reflects a slight drop in imports due to lower petroleum prices and more resilient remittances (Text Table 1). Exports of goods and services are expected to increase significantly by 2025 (from 21 percent of GDP to 34 percent of GDP) driven by the onset of oil and gas exports which accounts for 3/4 of the increase.
- The debt service path reflects the reprofiling of **DSSI**-eligible debt in 2020 (May-December) and 2021 (January-June) on standard terms, except for the debt from creditors whose participation is still pending. This will result in slightly higher debt service over the medium term.
- Senegal's public debt profile has also benefited from favorable ***exchange rate*** movements due to the depreciation of the USD relative to the CFAF/euro peg since the previous DSA (by about 6 percent).
- A more precise calculation of the ***present value of debt service*** has resulted in a slightly lower debt service path than in the April DSA.

4. The DSA assumes a financing mix consistent with a prudent borrowing strategy, aimed at gradually increasing the share of domestic debt and seeking new external financing on concessional terms whenever feasible. Financing for COVID-19-related expenses will be filled mostly with grants and highly concessional loans. The authorities have also delayed some lower-priority projects in 2020. The prospects for domestic borrowing are further supported by the measures taken by the BCEAO to enhance liquidity in the sub-regional market. However, the lack of market depth, combined with potentially higher demand from other countries in the region, serves as a constraint on domestic borrowing.

RISK RATING AND VULNERABILITIES

5. Senegal remains at moderate risk of external and public debt distress. The current baseline remains below the thresholds throughout the forecast period, with the exception of a one-off breach in the debt service-to-exports indicator in 2020. However, under the most extreme shock scenarios, three of the four external debt indicators breach their thresholds for a significant period, pointing to potential vulnerabilities under adverse conditions. The public debt-to-GDP ratio is projected to peak at around 68 percent of GDP in 2021–22 before gradually declining over the medium term as oil and gas investments come online. This is broadly unchanged from the April DSA. The higher central government fiscal deficit and lower growth have been broadly offset by favorable exchange rate movements and updated public sector revenue and expenditure data. Total public debt would also breach the threshold under the most extreme stress test (growth shock) in the later years of the decade, which supports the view that Senegal

should maintain a prudent fiscal path. Debt service is projected to be about 21 percent of total revenues and grants in 2020, before gradually rising to about 24 percent on average over the medium term due to borrowing related to the hydrocarbon sector. As such, debt service would continue to absorb a significant portion of fiscal revenues. The DSA's realism tools do not signal any signs of over-optimism in fiscal projections or large changes to investment and contributions to growth compared to previous DSAs.

Text Table 2. Senegal: Evolution of Selected Macroeconomic Indicators, 2020–23

	2020	2021	2022	2023	Med. term[1]	Long term[2]
Real GDP growth						
Current DSA	-0.7	5.2	6.0	11.9	6.2	5.1
Previous DSA	3.0	5.5	8.0	11.2	6.8	5.1
Overall fiscal deficit (percent of GDP)[3]						
Current DSA	7.4	5.8	5.2	4.0	4.8	2.7
Previous DSA	6.9	4.6	4.3	4.1	4.6	3.4
Current account deficit (percent of GDP)						
Current DSA	10.9	10.7	10.6	4.6	7.4	1.3
Previous DSA	11.3	11.4	8.5	3.7	7.8	1.5
Exports of goods and services (percent of GDP)						
Current DSA	21.0	22.3	24.1	32.7	28.1	32.2
Previous DSA	20.9	22.9	26.2	32.9	26.7	31.4

[1] Defined as the first 5 years of the projection period. For the current DSA update, the medium term covers the years 2020-2025.

[2] Defined as the last 15 years of the projection period. For the current DSA update, the long term covers the years 2026-2040.

[3] Overall fiscal deficit of General Government and Public Sector.

6. Overall, these results suggest that Senegal has limited space to absorb shocks in the near term and reinforces the need for prudent fiscal policy and sound debt management. The relatively high debt service levels throughout the forecast period would limit fiscal space for additional expenditures to support a robust and resilient recovery. This calls for further efforts to mobilize additional revenues and prioritize concessional borrowing.

7. Risks to the outlook depend primarily on the depth and duration of the COVID-19 pandemic. A deeper global slowdown combined with a prolonged outbreak in Senegal or its major trading partners could further lower GDP and export receipts in 2020 and weaken the expected recovery thereafter. Regional security risks and natural disasters could also increase fiscal pressures, contributing to a higher debt path. Lower oil prices benefit Senegal in the short term and help mitigate the current account deterioration but, should world oil and gas prices remain low for a prolonged period, could hinder planned investments in the oil and gas sector, with significant knock-on implications for the medium-term outlook.

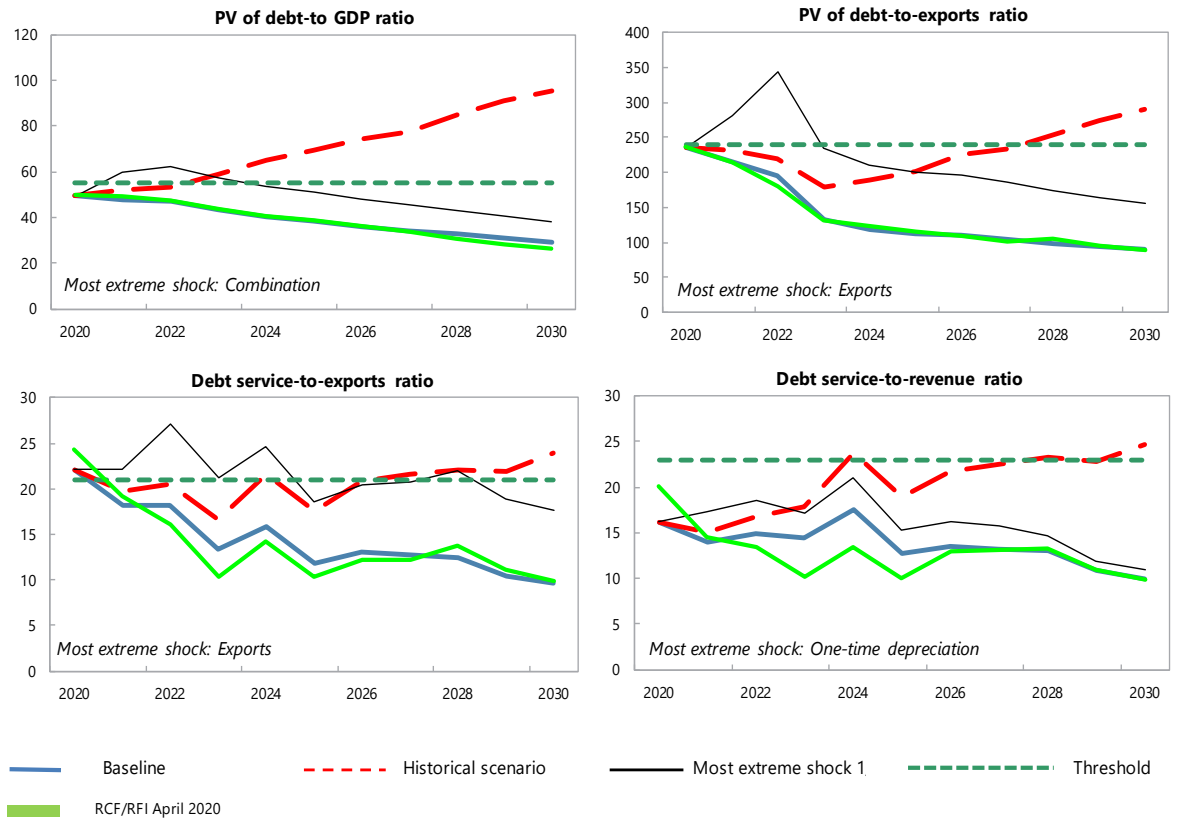
AUTHORITIES' VIEWS

8. The authorities broadly share staff's assessment of debt sustainability. They agree that risks to the debt profile are moderate with limited space to absorb shocks in the near term. Nevertheless, they are of the view that major investments under the PAP2A are needed to bolster the recovery and foster sustainable growth. To that end, they are developing a borrowing strategy for 2021–23 that places greater emphasis on concessional or semi-concessional external financing, as well as increased domestic borrowing to minimize foreign exchange risk. They are also seeking to strengthen oversight over public sector borrowing, including SOEs and PPPs, through the National Public Debt Committed (CNDP).

Text Table 3. Senegal: Calculation of the CI Score

Components	Coefficients (A)	10-year average values (B)	CI Score components (A*B) = (C)	Contribution of components
CPIA	0.385	3.753	1.45	45%
Real growth rate (in percent)	2.719	6.291	0.17	5%
Import coverage of reserves (in percent)	4.052	39.093	1.58	50%
Import coverage of reserves^2 (in percent)	-3.990	15.283	-0.61	-19%
Remittances (in percent)	2.022	10.261	0.21	6%
World economic growth (in percent)	13.520	2.928	0.40	12%
CI Score			3.19	100%
CI rating			Strong	

Figure 1. Senegal: Indicators of Public and Publicly Guaranteed External Debt Under Alternative Scenarios, 2020–30



Customization of Default Settings	Size	Interactions
Tailored Stress		
Combined CL	Yes	
Natural disaster	n.a.	n.a.
Commodity price 2/	n.a.	n.a.
Market financing	No	No

Note: "Yes" indicates any change to the size or interactions of the default settings for the stress tests. "n.a." indicates that the stress test does not apply.

Borrowing assumptions on additional financing needs resulting from the stress tests*	Default	User defined
Shares of marginal debt		
External PPG MLT debt	100%	
Terms of marginal debt		
Avg. nominal interest rate on new borrowing in USD	4.1%	4.1%
USD Discount rate	5.0%	5.0%
Avg. maturity (incl. grace period)	16	16
Avg. grace period	5	5

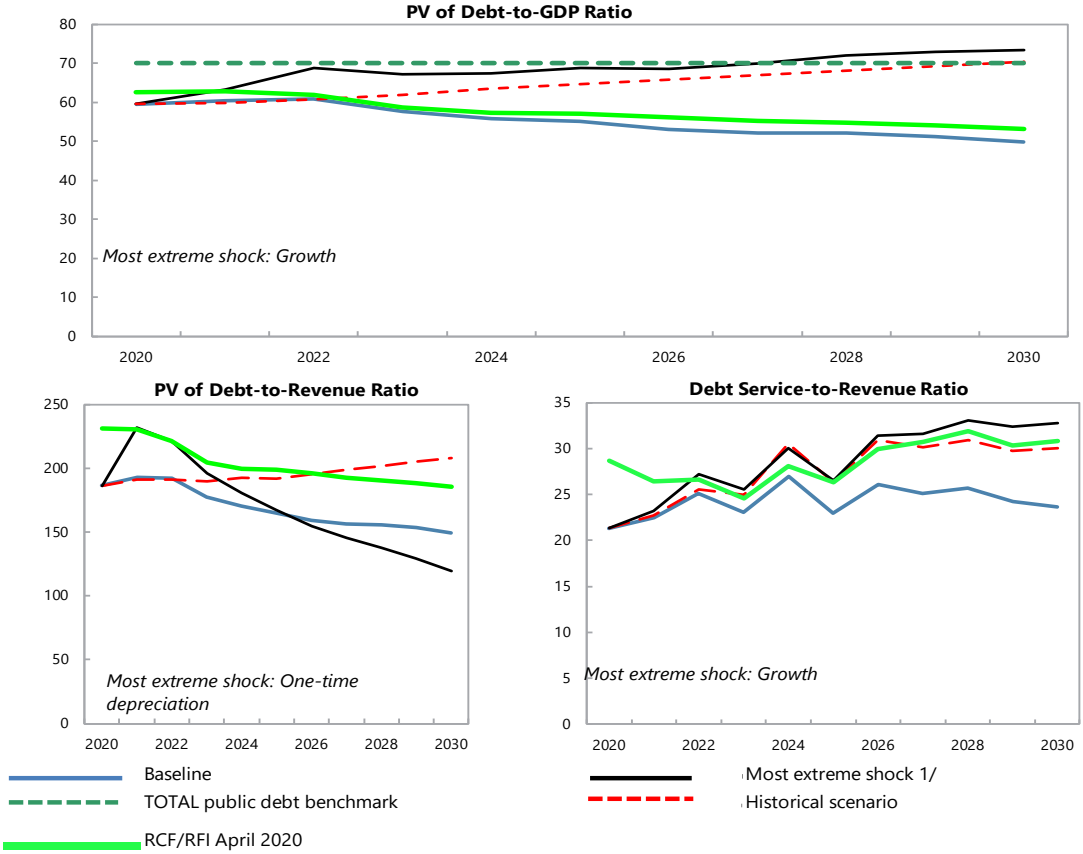
* Note: All the additional financing needs generated by the shocks under the stress tests are assumed to be covered by PPG external MLT debt in the external DSA. Default terms of marginal debt are based on baseline 10-year projections.

Sources: Country authorities; and staff estimates and projections.

1/ The most extreme stress test is the test that yields the highest ratio in or before 2030. The stress test with a one-off breach is also presented (if any), while the one-off breach is deemed away for mechanical signals. When a stress test with a one-off breach happens to be the most exterme shock even after disregarding the one-off breach, only that stress test (with a one-off breach) would be presented.

2/ The magnitude of shocks used for the commodity price shock stress test are based on the commodity prices outlook prepared by the IMF research department.

Figure 2. Senegal: Indicators of Public Debt Under Alternative Scenarios, 2020–30



Borrowing assumptions on additional financing needs resulting from the stress tests*	Default	User defined
Shares of marginal debt		
External PPG medium and long-term	47%	47%
Domestic medium and long-term	48%	48%
Domestic short-term	5%	5%
Terms of marginal debt		
External MLT debt		
Avg. nominal interest rate on new borrowing in USD	4.1%	4.1%
Avg. maturity (incl. grace period)	16	16
Avg. grace period	5	5
Domestic MLT debt		
Avg. real interest rate on new borrowing	4.0%	4.0%
Avg. maturity (incl. grace period)	5	5
Avg. grace period	3	3
Domestic short-term debt		
Avg. real interest rate	3.1%	3.1%

* Note: The public DSA allows for domestic financing to cover the additional financing needs generated by the shocks under the stress tests in the public DSA. Default terms of marginal debt are based on baseline 10-year projections.

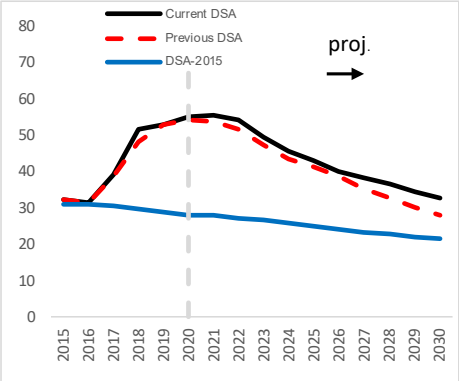
Sources: Country authorities; and staff estimates and projections.

1/ The most extreme stress test is the test that yields the highest ratio in or before 2030. The stress test with a one-off breach is also presented (if any), while the one-off breach is deemed away for mechanical signals. When a stress test with a one-off breach happens to be the most exterme shock even after disregarding the one-off breach, only that stress test (with a one-off breach) would be presented.

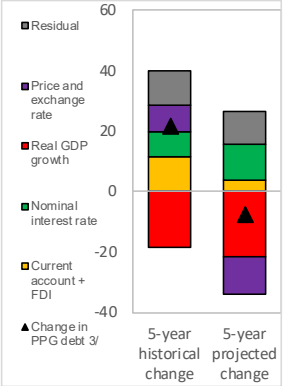
Figure 3. Senegal: Driver of Debt Dynamics—Baseline Scenario, 2015–30

External debt

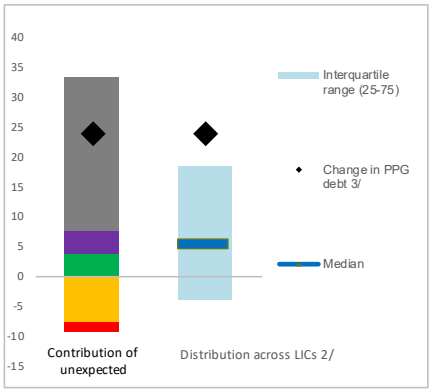
Gross Nominal PPG External Debt
(in percent of GDP; DSA vintages)



Debt-creating flows
(percent of GDP)

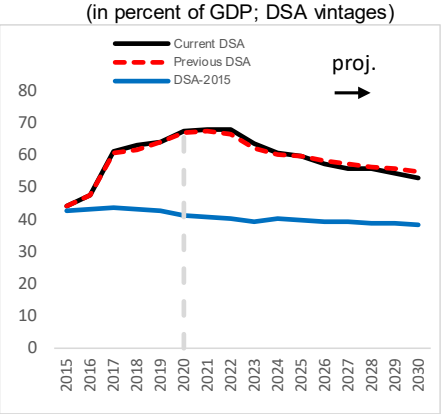


Unexpected Changes in Debt 1/
(past 5 years, percent of GDP)

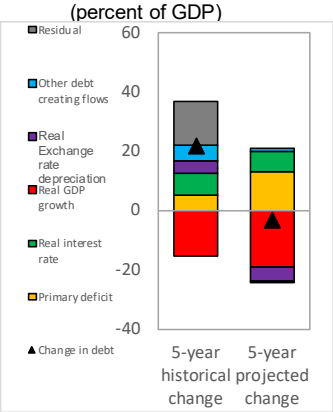


Public debt

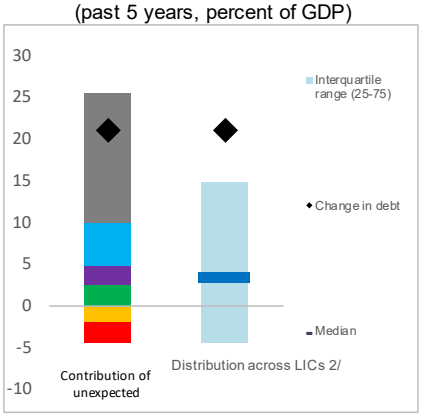
Gross Nominal Public Debt
(in percent of GDP; DSA vintages)



Debt-creating flows
(percent of GDP)



Unexpected Changes in Debt 1/
(past 5 years, percent of GDP)



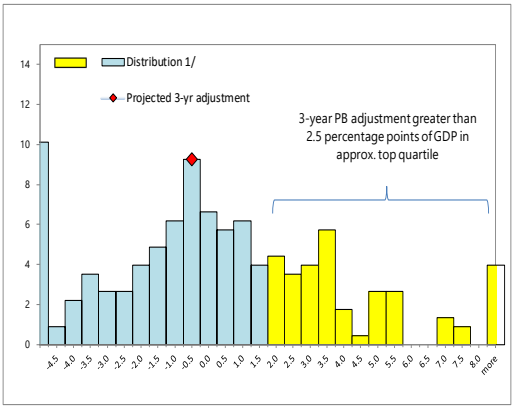
1/ Difference between anticipated and actual contributions on debt ratios.

2/ Distribution across LICs for which LIC DSAs were produced.

3/ Given the relatively low private external debt for average low-income countries, a ppt change in PPG external debt should be largely explained by the drivers of the external debt dynamics equation.

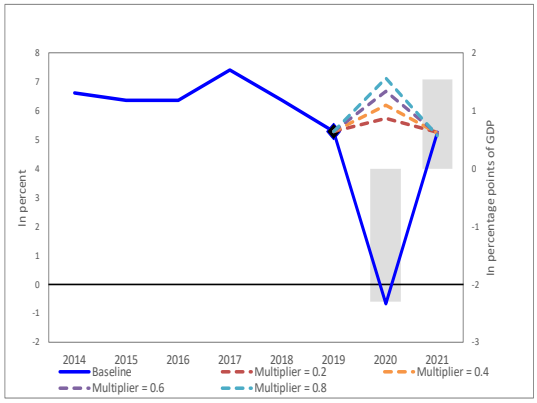
Figure 4. Senegal: Realism Tools, 2016–25

3-Year Adjustment in Primary Balance
(Percentage points of GDP)



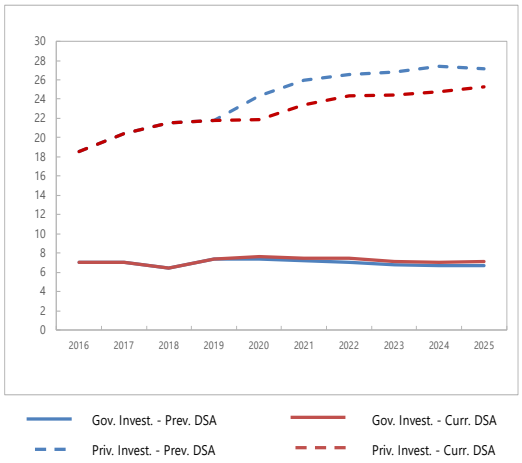
1/ Data cover Fund-supported programs for LICs (excluding emergency financing) approved since 1990. The size of 3-year adjustment from program inception is found on the horizontal axis; the percent of sample is found on the vertical axis.

Fiscal Adjustment and Possible Growth Paths 1/



1/ Bars refer to annual projected fiscal adjustment (right-hand side scale) and lines show possible real GDP growth paths under different fiscal multipliers (left-hand side scale).

Public and Private Investment Rates
(percent of GDP)



Contribution to Real GDP growth
(percent, 5-year average)

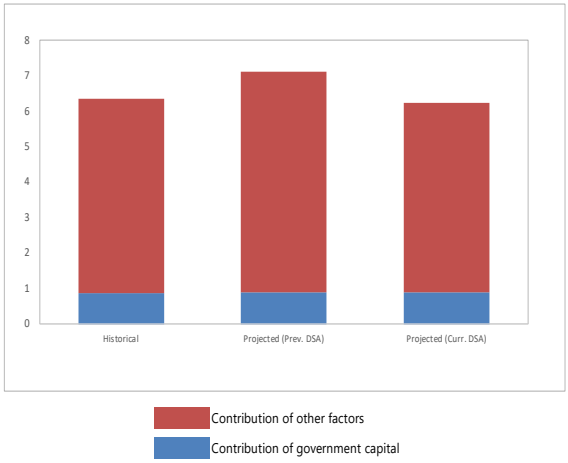
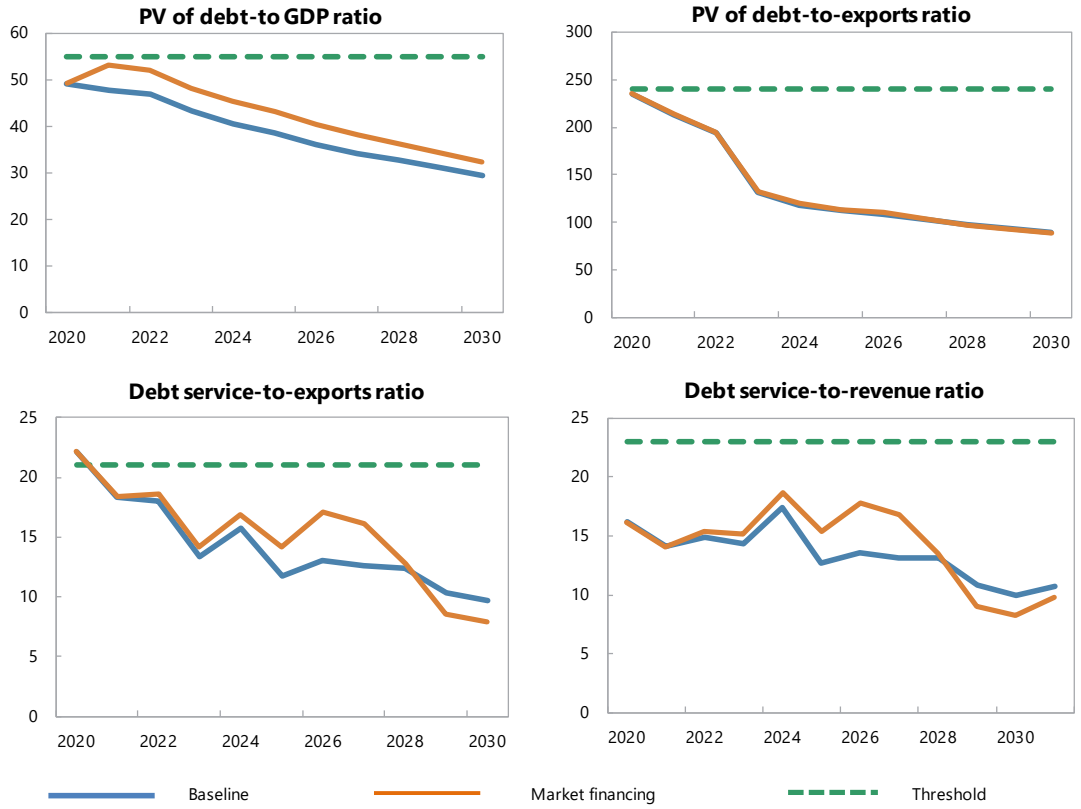


Figure 5. Senegal: Market-Financing Risk Indicators, 2020–30

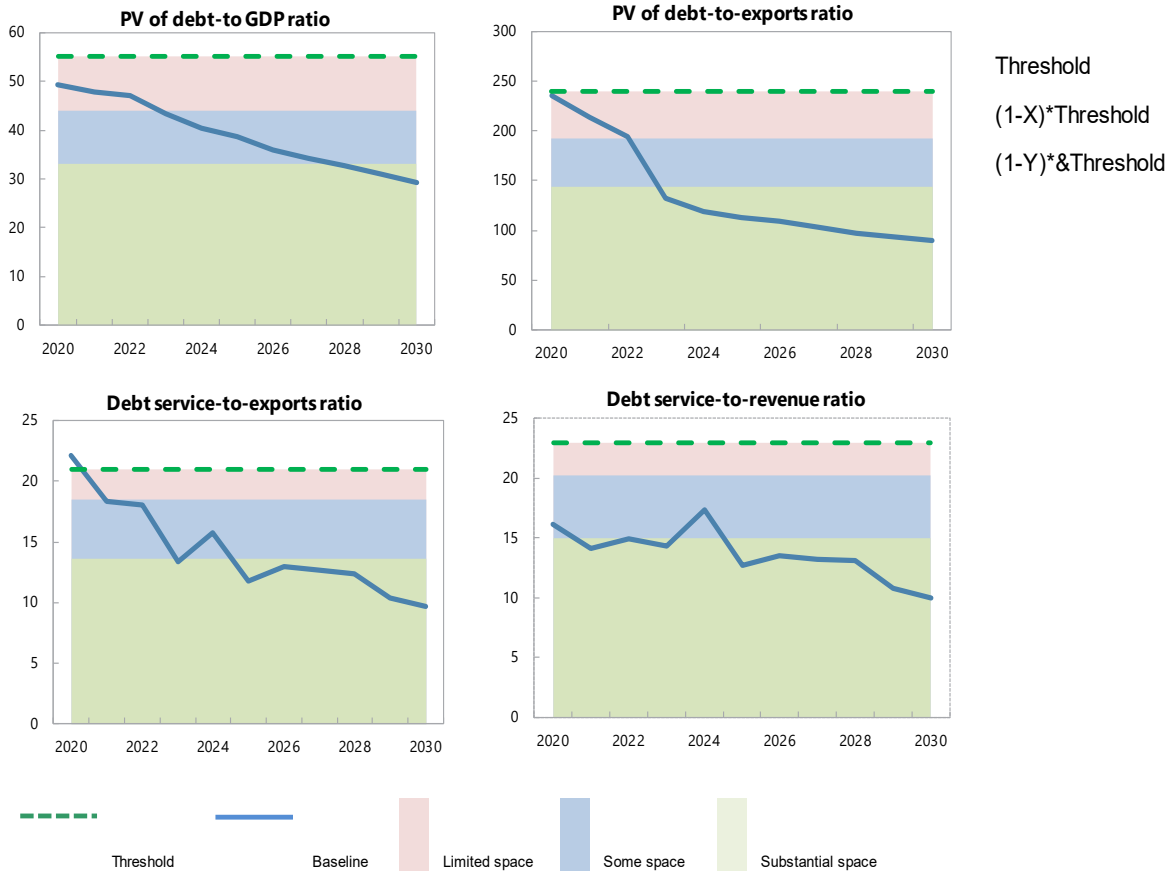
	GFN 1/	EMBI 2/
Benchmarks	14	570
Values	12	405
Breach of benchmark	No	No
Potential heightened liquidity needs	Low	

1/ Maximum gross financing needs (GFN) over 3-year baseline projection horizon.
2/ EMBI spreads correspond to the latest available data.



Sources: Country authorities; and staff estimates and projections.

Figure 6. Senegal: Qualification of the Moderate Category, 2020–30[1]



Sources: Country authorities; and staff estimates and projections.

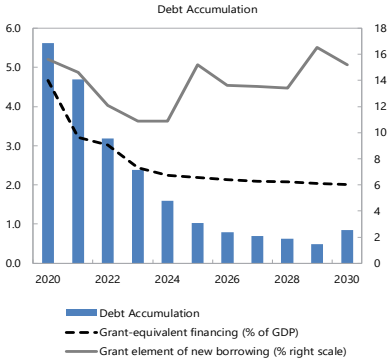
1/ For the PV debt/GDP and PV debt/exports thresholds, x is 20 percent and y is 40 percent. For debt service/Exports and debt service/revenue thresholds, x is 12 percent and y is 35 percent.

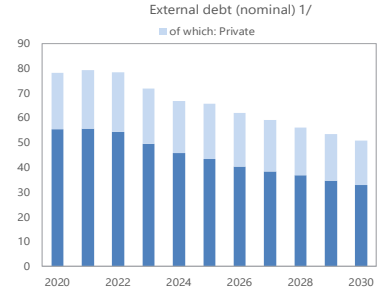
Table 1. Senegal: External Debt Sustainability Framework, Baseline Scenario, 2017–40

(percent of GDP unless otherwise indicated)

	Actual			Projections								Average 8/	
	2017	2018	2019	2020	2021	2022	2023	2024	2025	2030	2040	Historical	Projections
External debt (nominal) 1/	**61.7**	**71.6**	**73.8**	**78.1**	**79.1**	**78.3**	**71.8**	**66.6**	**65.5**	**50.7**	**36.6**	**56.1**	**65.5**
of which: public and publicly guaranteed (PPG)	**39.1**	**51.8**	**53.2**	**55.3**	**55.5**	**54.3**	**49.5**	**45.7**	**43.3**	**32.7**	**21.7**	**33.5**	**44.2**
Change in external debt	6.9	9.9	2.2	4.3	1.0	-0.8	-6.5	-5.1	-1.1	-2.6	-0.9		
Identified net debt-creating flows	**-0.2**	**-0.7**	**2.3**	**6.0**	**1.7**	**1.0**	**-8.1**	**-6.3**	**-3.9**	**-3.9**	**-4.9**	**2.8**	**-2.7**
Non-interest current account deficit	**6.0**	**6.6**	**5.4**	**8.8**	**8.2**	**8.1**	**2.2**	**1.1**	**1.7**	**0.1**	**-1.5**	**5.4**	**2.9**
Deficit in balance of goods and services	13.8	15.0	13.8	18.8	16.6	15.1	7.2	5.5	6.1	-0.5	-1.7	**14.2**	**6.7**
Exports	21.9	23.5	24.2	21.0	22.3	24.1	32.7	34.1	34.3	32.7	31.9		
Imports	35.7	38.6	38.0	39.8	39.0	39.2	39.9	39.6	40.4	32.2	30.3		
Net current transfers (negative = inflow)	-9.4	-8.9	-8.2	-10.3	-8.9	-9.1	-8.5	-8.1	-8.2	-7.8	-7.6	**-9.5**	**-8.4**
of which: official	-0.4	-0.3	0.1	-1.2	-0.3	-0.6	-0.3	-0.1	-0.1	-0.1	-0.3		
Other current account flows (negative = net inflow)	1.6	0.4	-0.2	0.3	0.5	2.1	3.4	3.7	3.8	8.5	7.8	**0.7**	**4.7**
Net FDI (negative = inflow)	**-2.4**	**-3.4**	**-4.2**	**-5.4**	**-5.4**	**-5.2**	**-4.5**	**-4.3**	**-4.4**	**-2.8**	**-2.7**	**-2.1**	**-4.2**
Endogenous debt dynamics 2/	**-3.9**	**-3.9**	**1.1**	**2.6**	**-1.1**	**-1.9**	**-5.8**	**-3.1**	**-1.2**	**-1.2**	**-0.7**		
Contribution from nominal interest rate	1.2	2.1	2.1	2.1	2.5	2.5	2.4	2.5	2.5	1.7	1.0		
Contribution from real GDP growth	-3.7	-3.5	-3.7	0.5	-3.6	-4.4	-8.2	-5.6	-3.7	-2.9	-1.7		
Contribution from price and exchange rate changes	-1.4	-2.5	2.7	...	...	...	...	...	...	...	...		
Residual 3/	**7.1**	**10.5**	**-0.1**	**-1.7**	**-0.7**	**-1.9**	**1.6**	**1.2**	**2.8**	**1.3**	**4.0**	**0.5**	**0.6**
of which: exceptional financing	0.6	1.1	-1.1	-0.2	-0.3	0.0	0.0	0.0	0.0	0.0	0.0		
Sustainability indicators													
PV of PPG external debt-to-GDP ratio	**...**	**...**	**45.1**	**49.2**	**47.9**	**47.0**	**43.3**	**40.5**	**38.6**	**29.4**	**19.2**		
PV of PPG external debt-to-exports ratio	**...**	**...**	**186.4**	**234.8**	**214.6**	**194.8**	**132.6**	**118.7**	**112.6**	**89.7**	**60.2**		
PPG debt service-to-exports ratio	**9.3**	**16.5**	**14.6**	**22.1**	**18.2**	**18.1**	**13.4**	**15.9**	**11.8**	**9.7**	**6.7**		
PPG debt service-to-revenue ratio	**7.5**	**14.0**	**12.1**	**16.1**	**13.9**	**14.9**	**14.4**	**17.5**	**12.8**	**10.0**	**6.7**		
Gross external financing need (Billion of U.S. dollars)	1.5	1.8	1.4	2.3	2.2	2.5	1.1	1.3	1.0	0.9	-1.7		
Key macroeconomic assumptions													
Real GDP growth (in percent)	7.4	6.4	5.3	-0.7	5.2	6.0	11.9	8.7	6.0	6.0	4.9	**5.1**	**5.9**
GDP deflator in US dollar terms (change in percent)	2.7	4.1	-3.7	3.8	6.9	2.5	1.9	2.1	1.5	2.3	2.1	**-1.1**	**2.7**
Effective interest rate (percent) 4/	2.5	3.8	3.0	2.9	3.6	3.4	3.5	3.9	4.0	3.5	2.9	**2.5**	**3.7**
Growth of exports of G&S (US dollar terms, in percent)	12.2	19.2	4.3	-10.7	19.9	17.4	54.4	15.8	8.1	7.0	8.3	**6.5**	**12.6**
Growth of imports of G&S (US dollar terms, in percent)	21.3	19.6	0.0	7.8	10.2	9.3	16.2	10.1	9.6	4.2	6.0	**6.0**	**7.2**
Grant element of new public sector borrowing (in percent)	...	...	...	15.6	14.6	12.1	10.9	10.9	15.2	15.2	18.0	**...**	**13.8**
Government revenues (excluding grants, in percent of GDP)	27.4	27.6	29.3	28.7	29.2	29.4	30.6	31.0	31.7	31.8	31.9	**20.1**	**30.8**
Aid flows (in Billion of US dollars) 5/	1.0	1.0	1.5	0.8	0.7	0.8	0.7	0.7	0.8	1.0	1.8		
Grant-equivalent financing (in percent of GDP) 6/	...	...	...	4.7	3.2	3.0	2.4	2.2	2.2	2.0	1.8	**...**	**2.6**
Grant-equivalent financing (in percent of external financing) 6/	...	...	...	37.5	35.6	37.7	37.5	34.1	44.2	45.4	51.8	**...**	**40.4**
Nominal GDP (Billion of US dollars)	21	23	24	24	27	30	34	38	40	59	116		
Nominal dollar GDP growth	10.3	10.8	1.4	3.1	12.5	8.6	14.0	11.0	7.6	8.4	7.1	**4.0**	**8.7**
Memorandum items:													
PV of external debt 7/	...	...	65.8	72.0	71.5	71.1	65.7	61.4	60.8	47.4	34.1		
In percent of exports	...	...	271.9	343.7	320.1	294.3	200.9	180.1	177.4	144.7	106.8		
Total external debt service-to-exports ratio	15.3	19.7	19.7	28.6	23.6	23.0	16.7	19.3	15.2	12.8	8.5		
PV of PPG external debt (in Billion of US dollars)			10.6	12.0	13.1	14.0	14.7	15.2	15.6	17.2	22.2		
(PVt-PVt-1)/GDPt-1 (in percent)				5.6	4.7	3.2	2.4	1.6	1.0	0.9	0.6		
Non-interest current account deficit that stabilizes debt ratio	-0.8	-3.2	3.2	4.5	7.2	9.0	8.7	6.2	2.8	2.7	-0.6		

Definition of external/domestic debt	Currency-based
Is there a material difference between the two criteria?	Yes





Sources: Country authorities; and staff estimates and projections.

1/ Includes both public and private sector external debt.

2/ Derived as [r - g - ρ(1+g)]/(1+g+ρ+gρ) times previous period debt ratio, with r = nominal interest rate; g = real GDP growth rate, and ρ = growth rate of GDP deflator in U.S. dollar terms.

3/ Includes exceptional financing (i.e., changes in arrears and debt relief); changes in gross foreign assets; and valuation adjustments. For projections also includes contribution from price and exchange rate changes.

4/ Current-year interest payments divided by previous period debt stock.

5/ Defined as grants, concessional loans, and debt relief.

6/ Grant-equivalent financing includes grants provided directly to the government and through new borrowing (difference between the face value and the PV of new debt).

7/ Assumes that PV of private sector debt is equivalent to its face value.

8/ Historical averages are generally derived over the past 10 years, subject to data availability, whereas projections averages are over the first year of projection and the next 10 years.

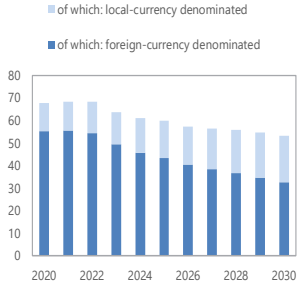
Table 2. Senegal: Public Sector Debt Sustainability Framework, Baseline Scenario, 2017–40

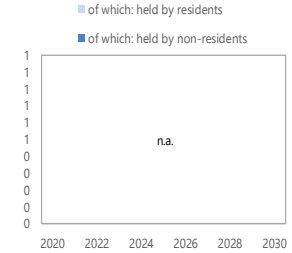
(percent of GDP, unless otherwise indicated)

	Actual			Projections								Average 6/	
	2017	2018	2019	2020	2021	2022	2023	2024	2025	2030	2040	Historical	Projections
Public sector debt 1/	**61.1**	**63.2**	**64.1**	**67.6**	**68.2**	**68.1**	**63.7**	**61.0**	**59.7**	**53.1**	**48.9**	**45.5**	**60.5**
of which: external debt	**39.1**	**51.8**	**53.2**	**55.3**	**55.5**	**54.3**	**49.5**	**45.7**	**43.3**	**32.7**	**21.7**	**33.5**	**44.2**
Change in public sector debt	13.6	2.1	0.9	3.5	0.7	-0.1	-4.4	-2.7	-1.3	-1.5	-0.2		
Identified debt-creating flows	**-6.7**	**3.0**	**2.6**	**7.2**	**1.4**	**0.2**	**-4.2**	**-2.7**	**-1.2**	**-1.5**	**-0.2**	**2.5**	**-0.5**
Primary deficit	**-2.8**	**1.8**	**2.4**	**4.7**	**3.2**	**2.7**	**1.5**	**1.1**	**0.6**	**0.4**	**0.8**	**2.1**	**1.4**
Revenue and grants	29.5	29.6	31.0	31.9	31.4	31.7	32.5	32.7	33.4	33.4	33.4	**22.1**	**32.8**
of which: grants	2.2	2.0	1.6	3.2	2.2	2.3	1.9	1.7	1.7	1.6	1.4		
Primary (noninterest) expenditure	26.7	31.4	33.4	36.6	34.5	34.4	34.0	33.8	34.0	33.8	34.2	**24.2**	**34.2**
Automatic debt dynamics	**-5.3**	**0.6**	**-0.6**	**1.6**	**-1.7**	**-2.5**	**-5.7**	**-3.8**	**-1.8**	**-1.9**	**-1.0**		
Contribution from interest rate/growth differential	-2.0	-0.9	-1.8	1.6	-1.7	-2.5	-5.7	-3.8	-1.8	-1.9	-1.0		
of which: contribution from average real interest rate	1.3	2.7	1.4	1.2	1.7	1.4	1.5	1.3	1.6	1.1	1.3		
of which: contribution from real GDP growth	-3.3	-3.7	-3.2	0.4	-3.4	-3.9	-7.2	-5.1	-3.4	-3.1	-2.3		
Contribution from real exchange rate depreciation	-3.3	1.5	1.2	...	...	...	...	...	...	...	...		
Other identified debt-creating flows	**1.4**	**0.6**	**0.7**	**0.9**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.6**	**0.1**
Privatization receipts (negative)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Recognition of contingent liabilities (e.g., bank recapitalization)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Debt relief (HIPC and other)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Other debt creating or reducing flow (please specify)	1.4	0.6	0.7	0.9	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Residual	**20.3**	**-0.9**	**-1.7**	**-3.7**	**-0.8**	**-0.3**	**-0.2**	**-0.1**	**-0.1**	**0.0**	**0.0**	**1.2**	**-0.5**
Sustainability indicators													
PV of public debt-to-GDP ratio 2/	**...**	**...**	**56.3**	**59.5**	**60.4**	**60.8**	**57.6**	**55.7**	**55.0**	**49.8**	**46.4**		
PV of public debt-to-revenue and grants ratio	**...**	**...**	**181.8**	**186.4**	**192.5**	**191.8**	**177.1**	**170.5**	**164.8**	**149.2**	**138.9**		
Debt service-to-revenue and grants ratio 3/	**23.1**	**31.9**	**21.7**	**21.3**	**22.5**	**25.1**	**23.1**	**27.0**	**22.9**	**23.7**	**23.9**		
Gross financing need 4/	5.4	11.8	9.8	12.4	10.2	10.6	9.0	9.9	8.2	8.3	8.8		
Key macroeconomic and fiscal assumptions													
Real GDP growth (in percent)	7.4	6.4	5.3	-0.7	5.2	6.0	11.9	8.7	6.0	6.0	4.9	**5.1**	**5.9**
Average nominal interest rate on external debt (in percent)	2.8	4.5	3.7	3.5	3.6	3.4	3.5	3.7	3.7	3.7	3.5	**2.9**	**3.6**
Average real interest rate on domestic debt (in percent)	4.2	4.2	3.9	4.1	5.8	4.3	4.4	4.1	4.5	3.7	3.9	**4.9**	**4.2**
Real exchange rate depreciation (in percent, + indicates depreciation)	-10.9	4.0	2.4	...	...	...	...	...	...	...	...	**3.0**	**...**
Inflation rate (GDP deflator, in percent)	0.6	-0.5	1.7	2.1	1.7	1.7	1.5	1.8	1.4	2.3	2.1	**0.9**	**1.9**
Growth of real primary spending (deflated by GDP deflator, in percent)	28.8	24.8	12.0	9.0	-0.8	5.5	10.6	8.0	6.5	5.9	5.0	**10.8**	**6.0**
Primary deficit that stabilizes the debt-to-GDP ratio 5/	-16.4	-0.3	1.6	1.2	2.5	2.8	5.9	3.8	1.8	1.9	1.0	**-5.0**	**2.4**
PV of contingent liabilities (not included in public sector debt)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		

Definition of external/domestic debt	**Currency-based**
Is there a material difference between the two criteria?	**Yes**

Public sector debt 1/





Sources: Country authorities; and staff estimates and projections.

1/ Coverage of debt: The central, state, and local governments plus extra budgetary funds, central bank, government-guaranteed debt, non-guaranteed SOE debt . Definition of external debt is Currency-based.

2/ The underlying PV of external debt-to-GDP ratio under the public DSA differs from the external DSA with the size of differences depending on exchange rates projections.

3/ Debt service is defined as the sum of interest and amortization of medium and long-term, and short-term debt.

4/ Gross financing need is defined as the primary deficit plus debt service plus the stock of short-term debt at the end of the last period and other debt creating/reducing flows.

5/ Defined as a primary deficit minus a change in the public debt-to-GDP ratio ((-): a primary surplus), which would stabilizes the debt ratio only in the year in question.

6/ Historical averages are generally derived over the past 10 years, subject to data availability, whereas projections averages are over the first year of projection and the next 10 years.

Table 3. Senegal: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, 2020–30

(percent)

	Projections 1/										
	2020	2021	2022	2023	2024	2025	2026	2027	2028	2029	2030
PV of debt-to GDP ratio											
Baseline	49	48	47	43	40	39	36	34	33	31	29
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2020-2030 2/	49	52	53	**59**	**65**	**69**	**74**	**77**	**85**	**91**	**95**
A2. Alternative Scenario :[Customize, enter title]	50	49	47	44	41	39	36	33	31	28	26
B. Bound Tests											
B1. Real GDP growth	49	49	51	47	44	42	39	37	35	33	32
B2. Primary balance	49	49	49	45	42	41	38	36	35	33	31
B3. Exports	49	52	**58**	53	50	48	45	42	40	38	35
B4. Other flows 3/	49	52	**56**	51	48	46	43	41	39	36	34
B5. Depreciation	49	**60**	52	48	45	43	40	38	36	35	33
B6. Combination of B1-B5	49	**60**	**62**	**57**	53	51	48	45	43	40	38
C. Tailored Tests											
C1. Combined contingent liabilities	49	51	50	46	43	42	40	38	36	35	33
C2. Natural disaster	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	49	53	52	48	45	43	40	38	36	34	32
Threshold	55	55	55	55	55	55	55	55	55	55	55
PV of debt-to-exports ratio											
Baseline	235	215	195	133	119	113	109	104	98	93	90
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2020-2030 2/	235	232	220	179	190	202	225	234	**253**	**274**	**291**
A2. Alternative Scenario :[Customize, enter title]	235	219	199	143	132	125	123	113	102	94	85
B. Bound Tests											
B1. Real GDP growth	235	215	195	133	119	113	109	104	98	93	90
B2. Primary balance	235	218	204	139	124	118	116	110	104	100	96
B3. Exports	235	**281**	**345**	234	210	200	195	185	173	164	155
B4. Other flows 3/	235	234	230	156	140	133	130	123	115	109	104
B5. Depreciation	235	215	174	119	106	101	97	92	87	84	81
B6. Combination of B1-B5	235	**290**	232	211	190	181	176	166	156	147	140
C. Tailored Tests											
C1. Combined contingent liabilities	235	228	208	141	127	122	120	115	108	104	101
C2. Natural disaster	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	235	215	195	133	120	114	110	104	98	93	89
Threshold	240	240	240	240	240	240	240	240	240	240	240
Debt service-to-exports ratio											
Baseline	**22**	18	18	13	16	12	13	13	12	10	10
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2020-2030 2/	**22**	20	20	17	**22**	18	21	**22**	**22**	**22**	**24**
A2. Alternative Scenario :[Customize, enter title]	**22**	19	19	15	19	14	16	15	14	11	10
B. Bound Tests											
B1. Real GDP growth	**22**	18	18	13	16	12	13	13	12	10	10
B2. Primary balance	**22**	18	18	14	16	12	13	13	13	11	10
B3. Exports	**22**	**22**	**27**	**21**	**25**	19	20	21	**22**	19	18
B4. Other flows 3/	**22**	18	19	14	17	13	14	14	15	13	12
B5. Depreciation	**22**	18	18	13	15	11	13	12	11	9	9
B6. Combination of B1-B5	**22**	**21**	**26**	20	**23**	17	19	20	20	17	16
C. Tailored Tests											
C1. Combined contingent liabilities	**22**	18	19	14	16	12	13	13	13	11	10
C2. Natural disaster	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	**22**	18	19	14	17	14	17	16	13	9	8
Threshold	21	21	21	21	21	21	21	21	21	21	21
Debt service-to-revenue ratio											
Baseline	16	14	15	14	17	13	14	13	13	11	10
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2020-2030 2/	16	15	17	18	**24**	19	22	22	**23**	23	**25**
A2. Alternative Scenario :[Customize, enter title]	16	14	16	16	21	15	17	16	15	12	11
B. Bound Tests	16	14	16	16	21	15	17	16	15	12	11
B1. Real GDP growth	16	14	16	15	19	14	15	14	14	12	11
B2. Primary balance	16	14	15	15	18	13	14	14	14	11	11
B3. Exports	16	14	15	16	19	14	15	15	16	14	13
B4. Other flows 3/	16	14	16	15	18	14	14	15	15	13	12
B5. Depreciation	16	17	19	17	21	15	16	16	15	12	11
B6. Combination of B1-B5	16	15	18	17	21	15	16	17	17	15	14
C. Tailored Tests											
C1. Combined contingent liabilities	16	14	15	15	18	13	14	14	14	11	10
C2. Natural disaster	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	16	14	15	15	19	15	18	17	14	9	8
Threshold	23	23	23	23	23	23	23	23	23	23	23

Sources: Country authorities; and staff estimates and projections.
1/ A bold value indicates a breach of the threshold.
2/ Variables include real GDP growth, GDP deflator (in U.S. dollar terms), non-interest current account in percent of GDP, and non-debt creating flows.
3/ Includes official and private transfers and FDI.

Table 4. Senegal: Sensitivity Analysis for Key Indicators of Public Debt, 2020–30

(Percent of GDP, unless otherwise indicated)

	Projections 1/										
	2020	2021	2022	2023	2024	2025	2026	2027	2028	2029	2030
PV of Debt-to-GDP Ratio											
Baseline	60	60	61	58	56	55	53	52	52	51	50
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2020-2030 2/	60	60	61	62	63	65	66	67	68	69	**70**
A2. Alternative Scenario :[Customize, enter title]	63	63	62	59	57	57	56	55	55	54	53
B. Bound Tests											
B1. Real GDP growth	60	63	69	67	67	69	69	70	**72**	**73**	**73**
B2. Primary balance	60	62	65	61	59	58	56	55	55	54	53
B3. Exports	60	64	**71**	67	65	64	61	60	59	58	56
B4. Other flows 3/	60	65	69	65	63	62	60	59	58	56	54
B5. Depreciation	60	**72**	70	63	59	56	51	48	46	43	40
B6. Combination of B1-B5	60	59	62	58	56	55	53	52	52	50	49
C. Tailored Tests											
C1. Combined contingent liabilities	60	67	67	64	62	61	58	57	57	56	55
C2. Natural disaster	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	60	60	61	58	56	55	53	52	52	51	50
TOTAL public debt benchmark	70	70	70	70	70	70	70	70	70	70	70
PV of Debt-to-Revenue Ratio											
Baseline	186	192	192	177	171	165	159	156	156	153	149
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2020-2030 2/	186	191	191	189	192	192	195	199	202	205	208
A2. Alternative Scenario :[Customize, enter title]	21	24	26	24	22	16	25	24	24	21	21
B. Bound Tests											
B1. Real GDP growth	186	201	215	206	205	205	205	209	215	218	219
B2. Primary balance	186	198	205	189	181	175	169	166	165	163	158
B3. Exports	186	204	225	207	198	191	184	180	178	173	166
B4. Other flows 3/	186	206	219	201	193	186	179	176	173	169	163
B5. Depreciation	186	232	221	196	181	167	154	145	138	129	119
B6. Combination of B1-B5	186	190	196	180	173	166	160	156	155	151	146
C. Tailored Tests											
C1. Combined contingent liabilities	186	214	213	196	188	182	175	172	172	169	164
C2. Natural disaster	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	186	192	192	178	172	166	160	157	156	153	148
Debt Service-to-Revenue Ratio											
Baseline	21	23	25	23	27	23	26	25	26	24	24
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2020-2030 2/	21	23	26	25	31	26	31	30	31	30	30
A2. Alternative Scenario :[Customize, enter title]	21	24	26	24	22	16	25	24	24	21	21
B. Bound Tests											
B1. Real GDP growth	21	23	27	26	30	27	31	32	33	32	33
B2. Primary balance	21	23	26	24	28	24	29	27	27	25	25
B3. Exports	21	23	26	24	28	24	27	27	28	27	26
B4. Other flows 3/	21	23	26	24	28	24	27	27	28	26	26
B5. Depreciation	21	24	29	27	31	26	28	27	28	25	24
B6. Combination of B1-B5	21	22	26	24	28	24	27	26	26	24	24
C. Tailored Tests											
C1. Combined contingent liabilities	21	23	27	24	28	27	31	27	27	26	26
C2. Natural disaster	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	21	23	26	24	28	25	30	29	26	22	22

Sources: Country authorities; and staff estimates and projections.

1/ A bold value indicates a breach of the benchmark.

2/ Variables include real GDP growth, GDP deflator and primary deficit in percent of GDP.

3/ Includes official and private transfers and FDI.